older physicists had similarly argued against Einstein's theory of relativity a generation earlier). Famously protesting that "God does not play dice," Einstein had a series of informal debates with Bohr, beginning at Solvay in 1927, in which he repeatedly tried to find a way around Heisenberg's uncertainty principle. Ultimately, Bohr prevailed, and the rest of the physics community accepted that quantum physics was correct and that the Copenhagen interpretation was the correct way to understand it. But Einstein never accepted the new theory, and until the day he died he insisted that nature could not be fundamentally random. Thus, the fable concludes, even the greatest and most famous physicists can still be wrong.



Figure 3.1. The Fifth Solvay Conference, Brussels, 1927. Front row: Einstein, center; Curie, third from left; Planck, second from left. Second row: Bohr, far right; Born, second from right; de Broglie, third from right. Back row: Heisenberg, third from right; Pauli, fourth from right; Schrödinger, center.

Some of this story is true. It's true that Einstein and Bohr disagreed about quantum physics. It's true that they debated it at the Solvay conference in 1927 and afterward. And it's true that Einstein said that "God does not play dice," though he said it in a letter to Max Born in 1926, not in Brussels in 1927. But in almost every other important respect—Einstein's real problem with quantum physics, Bohr's defense of it, even the content of the Copenhagen interpretation and its general acceptance by the rest of